

Deep Learning for Solving Dynamic Models

Session 8: Economics-informed neural networks, foundations

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Based on Raissi, Perdikaris & Karniadakis (2019), *Journal of Computational Physics* 378, 686–707

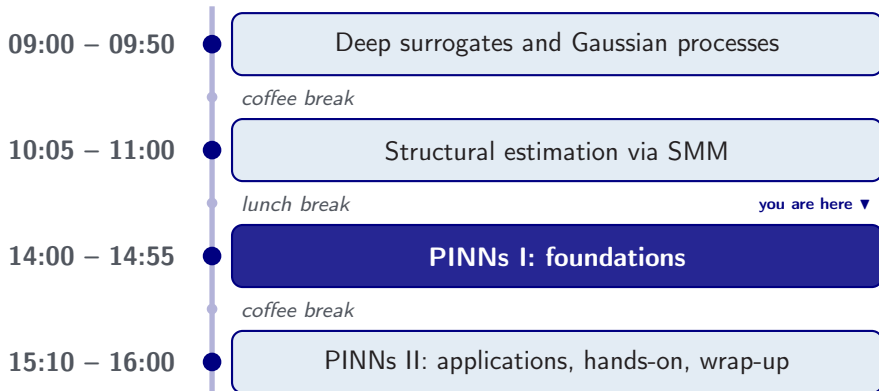
University of Padova

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Further: Sirignano & Spiliopoulos (2018), Deep Galerkin Method for PDEs in high dimensions.

Al-Aradi, Correia, Naez, Hoshisashi & Schwab (2018), DGM PDE solver.

Day 2 at a Glance



Outline

1. From DEQNs to PINNs

Bridging discrete and continuous time; the PINN loss.

2. PINN Fundamentals

Automatic differentiation for PDEs, collocation, training loop.

3. Soft vs. Hard Boundary Conditions

Trial solutions, 2D Poisson, practical guidance.

4. Economic Applications: HJB Equation

Cake-eating problem, scaled trial solution + hard BCs, Adam $\rightarrow$ L-BFGS, analytical benchmark; DGM as optional high-D architecture.

5. Financial Applications: Black–Scholes

Option pricing via PINNs, comparison with analytical formula.

6. When to Use One, and What Goes Wrong

PINNs vs. grid solvers, inverse problems, failure modes, and which notebooks to run.

Roadmap of This Session (55 min)

Part I of two. This session builds the method; Session 9 after the break applies it to economic and financial models.

I. From DEQNs to PINNs (12 min)

- ▶ Discrete vs. continuous time, the PINN loss
- ▶ Finger exercise

II. PINN fundamentals (20 min)

- ▶ Autograd for PDEs, collocation, the training loop
- ▶ Adam $\rightarrow$ L-BFGS in FP64; warm-up 1D ODE

III. Boundary conditions (23 min)

- ▶ Soft penalties vs. hard trial solutions
- ▶ Hard BCs in 2D; the Poisson equation; finger exercise

Part I

From DEQNs to PINNs

Recall: The DEQN Loss from Day 1

Yesterday morning (02_DEQN.pdf) **we trained** a neural network $\mathcal{N}_\theta$ to satisfy economic equilibrium conditions in **discrete time**:

$$\ell_\theta^{\text{DEQN}} = \frac{1}{N} \sum_{i=1}^N \|G(\mathbf{x}_i, \mathcal{N}_\theta(\mathbf{x}_i))\|^2$$

- ▶ $G(\cdot) = 0$: Euler equations, budget constraints, market clearing
- ▶ Training points $\mathbf{x}_i$: simulated from the model's own dynamics
- ▶ **No labeled data needed**: structural equations *are* the loss

Today's question: What if the economic model is formulated in **continuous time**?

Discrete vs. Continuous Time

Discrete Time (DEQN)

Euler equation:

$$u'(c_t) = \beta \mathbb{E}[u'(c_{t+1}) R_{t+1}]$$

State transition:

$$\mathbf{x}_{t+1} = h(\mathbf{x}_t, p(\mathbf{x}_t), \varepsilon_{t+1})$$

Loss: algebraic residual $G(\cdot) = 0$

Continuous Time (PINN)

State evolution:

$$da = (ra - c) dt + \sigma dW$$

HJB equation:

$$\rho V = \max_c \{ u(c) + V_a(ra - c) + \frac{1}{2}\sigma^2 V_{aa} \}$$

Loss: PDE residual $\mathcal{R}(\cdot) = 0$

Key shift: from algebraic residuals to **differential** residuals, we need **derivatives** of the neural network. The diffusion term $\frac{1}{2}\sigma^2 V_{aa}$ is why **second** derivatives matter.

Why Continuous Time Matters

Many frontier models are formulated in continuous time:

- ▶ **Heterogeneous-agent models:** Achdou, Han, Lasry, Lions & Moll (2022), income and wealth distribution dynamics via HJB–KFE systems
- ▶ **Mean-field games:** optimal policy with a continuum of interacting agents
- ▶ **Asset pricing:** Black–Scholes, Heston, jump-diffusion models
- ▶ **Macro-finance:** Brunnermeier & Sannikov (2014), He & Krishnamurthy (2013)

Where this sits in the course

Day 1 solved equilibrium conditions in **discrete** time with DEQNs. This afternoon we do the **continuous-time** counterpart. Coupling the HJB to a Kolmogorov forward equation, for heterogeneous agents, is the optional material in `day2/optional_continuous_time_ha`.

The PINN Loss Function

Core idea (Raissi, Perdikaris & Karniadakis, 2019):

Given a PDE $\mathcal{D}[u](\mathbf{x}) = 0$ on domain Ω with boundary conditions $\mathcal{B}[u](\mathbf{x}) = 0$ on $\partial\Omega$, define:

$$\ell_{\theta}^{\text{PINN}} = \underbrace{\frac{1}{N_r} \sum_{i=1}^{N_r} |\mathcal{D}[\mathcal{N}_{\theta}](\mathbf{x}_i^r)|^2}_{\text{PDE residual}} + \underbrace{\frac{\lambda}{N_b} \sum_{j=1}^{N_b} |\mathcal{B}[\mathcal{N}_{\theta}](\mathbf{x}_j^b)|^2}_{\text{boundary/initial conditions}}$$

- ▶ $\mathcal{N}_{\theta}$: neural network with parameters θ
- ▶ $\mathbf{x}_i^r$: **collocation points** in the interior of Ω
- ▶ $\mathbf{x}_j^b$: points on the boundary $\partial\Omega$
- ▶ Derivatives $\mathcal{D}[\mathcal{N}_{\theta}]$ computed via **automatic differentiation**

Warm-up ODE: for $u''(x) = -1$ on $(0, 1)$ with $u(0) = u(1) = 0$, use $\mathcal{D}[u] = u'' + 1$ in the interior and $\mathcal{B}[u] = (u(0), u(1))$ on the boundary.

Finger Exercise: Writing a PINN Loss

Exercise (5 min)

Consider the ODE $u''(x) = -1$ on $[0, 1]$ with boundary conditions $u(0) = u(1) = 0$.

1. Write down the PINN residual $\mathcal{R}(x; \theta) = \dots$ using a neural network $\mathcal{N}(x; \theta)$.
2. Write the total PINN loss as a sum of a PDE residual term and a boundary term.
3. What is the exact solution?

Solution: Writing a PINN Loss

Answer

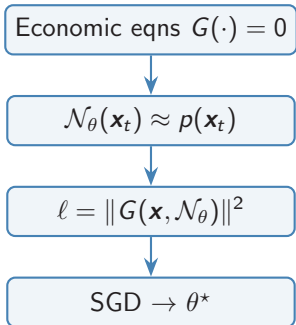
1. $\mathcal{R}(x; \theta) = \frac{\partial^2 \mathcal{N}}{\partial x^2}(x; \theta) + 1.$

2. $\mathcal{L}(\theta) = \underbrace{\frac{1}{N_r} \sum_{i=1}^{N_r} |\mathcal{R}(x_i; \theta)|^2}_{\text{PDE residual}} + \lambda \underbrace{[\mathcal{N}(0; \theta)^2 + \mathcal{N}(1; \theta)^2]}_{\text{boundary conditions}}$

3. $u(x) = \frac{1}{2}x(1-x)$ satisfies $u'' = -1$, $u(0) = u(1) = 0$. ✓

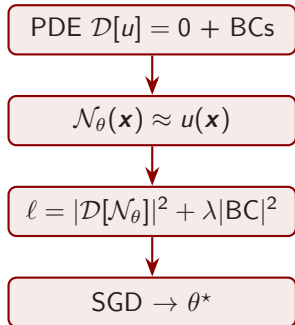
DEQN vs. PINN: Side-by-Side

DEQN (Discrete Time)



← same principle →
+ autograd derivatives

PINN (Continuous Time)



Both approaches are **unsupervised**: the physics (economics) is the loss function. The key addition for PINNs: we differentiate the network via any auto-diff tool (e.g. `torch.autograd.grad`).

Part II

PINN Fundamentals

Automatic Differentiation for PDEs

Key tool: `torch.autograd.grad` with `create_graph=True`.

```
# First derivative dy/dx  
dydx = torch.autograd.grad(  
    y, x,  
    grad_outputs=torch.ones_like(y),  
    create_graph=True # keep graph for higher derivs  
)[0]
```

- ▶ `create_graph=True`: retains the computational graph so we can differentiate **again**
- ▶ This gives us **algorithmic** derivatives (up to floating-point precision), no finite differences!
- ▶ Works for **any order**: $\frac{\partial^n}{\partial x^n}$ by chaining n calls

Computing Second Derivatives

Chain two autograd calls to get $y''(x)$:

First derivative

```
dydx = torch.autograd.grad(  
    y, x, grad_outputs=torch.ones_like(y),  
    create_graph=True)[0]
```

Second derivative

```
d2ydx2 = torch.autograd.grad(  
    dydx, x, grad_outputs=torch.ones_like(dydx),  
    create_graph=True)[0]
```



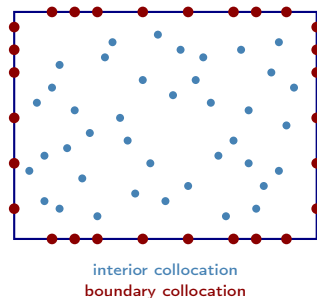
Collocation Points

PINNs evaluate the PDE residual at **randomly sampled** points, no grid required.

1D: interior + boundary



2D: $\Omega = (0, 1)^2$



Points are **resampled** periodically during training to avoid overfitting to a fixed set.

PINN Training Algorithm

Algorithm: Physics-Informed Neural Network Training

Input: PDE $\mathcal{D}[u] = 0$, BCs $\mathcal{B}[u] = 0$, network $\mathcal{N}_\theta$, learning rate η

for epoch = $1, \dots, E$ **do**

 Sample N_r collocation points $\{\mathbf{x}_i^r\}$ in domain Ω

 Sample N_b boundary points $\{\mathbf{x}_j^b\}$ on $\partial\Omega$

 Forward pass: compute $\mathcal{N}_\theta(\mathbf{x}_i^r)$ and $\mathcal{N}_\theta(\mathbf{x}_j^b)$

 Autograd: compute $\mathcal{D}[\mathcal{N}_\theta](\mathbf{x}_i^r)$ via `torch.autograd.grad`

$\ell_\theta = \frac{1}{N_r} \sum_i |\mathcal{D}[\mathcal{N}_\theta](\mathbf{x}_i^r)|^2 + \frac{\lambda}{N_b} \sum_j |\mathcal{B}[\mathcal{N}_\theta](\mathbf{x}_j^b)|^2$

 Update: $\theta \leftarrow \theta - \eta \nabla_\theta \ell_\theta$

end for

Output: Trained $\mathcal{N}_{\theta^*}$ approximating the PDE solution

Note: Unlike DEQNs, training points are **not** simulated from model dynamics, they are drawn uniformly (or adaptively) from the domain.

Optimizer Pipeline: Adam $\rightarrow$ L-BFGS, in FP64

Two stages.

- ▶ **Adam** on resampled collocation: finds a basin.
- ▶ **Deterministic L-BFGS** on a *fixed* grid: closes the residual. Resampling per evaluation breaks the line search.

Why FP64?

- ▶ tanh-MLP second derivatives lose ~ 7 digits in FP32.
- ▶ The Wolfe search compares values at exactly that scale.
- ▶ FP32 plateaus at the noise floor; FP64 keeps going.

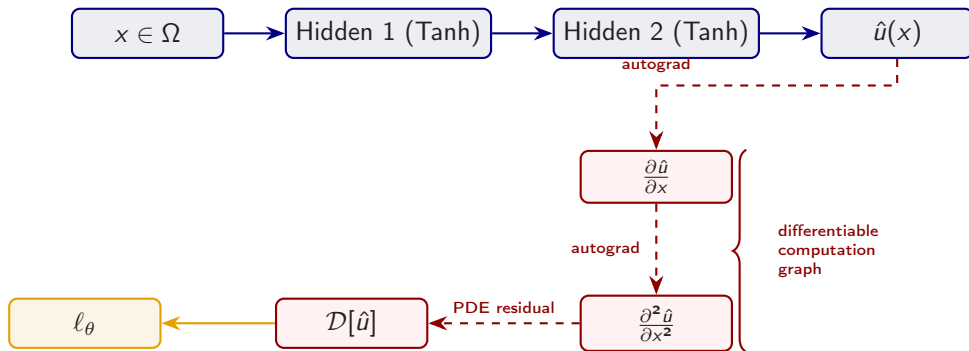
Used in notebooks 04 and 05. Report PDE residuals and solution diagnostics, not the optimizer label.

Practical recipe

1. Cast model + data to float64.
2. Adam, lr 10^{-3} , a few k epochs, resampled collocation.
3. Freeze grid; `torch.optim.LBFGS` (strong-Wolfe).
4. Stop on residual stagnation.

Precision evidence: [arXiv:2505.10949](#).

PINN Architecture



The entire pipeline, forward pass *and* derivative computation, is **differentiable**, enabling end-to-end gradient-based training.

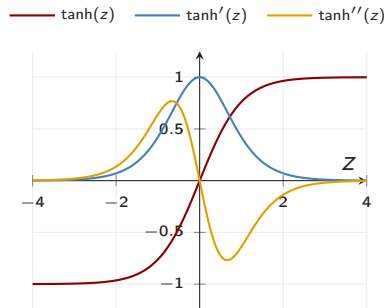
Why Tanh Activations for PINNs?

PDE residuals require smooth derivatives.

- ▶ **ReLU**: C^0 only; its second derivative vanishes almost everywhere, so a u'' residual carries no gradient
- ▶ **Tanh**: C^∞ , all derivatives are well-defined and informative
- ▶ **Softplus**: C^∞ , useful for positivity constraints (e.g. consumption)

Rule of thumb

For PINNs requiring k -th order derivatives, use activations that are at least C^k .



Warm-up: Solving an ODE with a PINN

Problem: $y''(x) = -1$ on $(0, 1)$, $y(0) = y(1) = 0$. **Analytical:** $y(x) = \frac{1}{2}x(1 - x)$.

PINN setup:

1. NN $\mathcal{N}_\theta : [0, 1] \rightarrow \mathbb{R}$, 2 hidden layers of 20 units, tanh
2. Residual: $r(x) = \mathcal{N}_\theta''(x) + 1$
3. Loss: PDE residual + BC penalty (soft, here)

$$\ell = \frac{1}{N} \sum_i r(x_i)^2 + \mathcal{N}_\theta(0)^2 + \mathcal{N}_\theta(1)^2$$

After 5,000 Adam epochs:

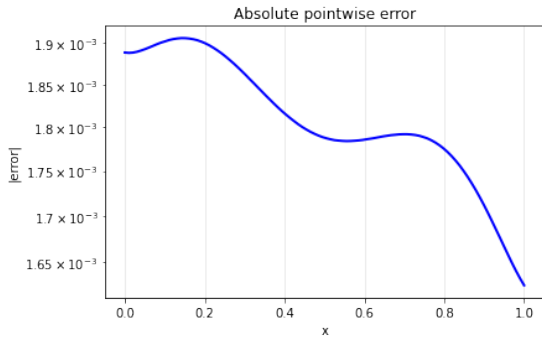
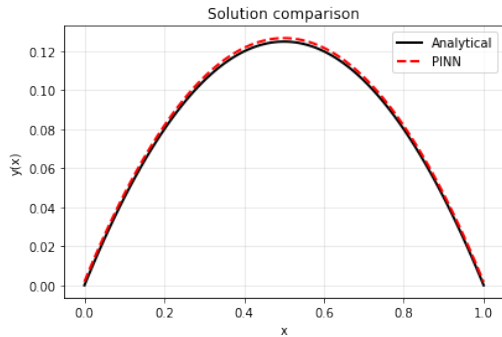
- ▶ Final loss $\approx 2 \times 10^{-5}$, of which the BC term is $\approx 9 \times 10^{-6}$
- ▶ Max absolute error 1.9×10^{-3} , mean 1.8×10^{-3}

Read the error, not the loss

A small loss says the residual is small at the sampled points. Only the comparison against $\frac{1}{2}x(1 - x)$ says the *solution* is right.

Notebook: 01_ODE_PINN_ZeroBCs

Warm-up: PINN vs. Analytical Solution



Left: the network on top of $y(x) = \frac{1}{2}x(1-x)$, indistinguishable at this scale. Right: the absolute error, which is what actually tells you how well it did. Note the axis: it stays in a narrow band around 1.8×10^{-3} across the domain, with no drift to the boundaries.

Notebook: 01_ODE_PINN_ZeroBCs

Non-Zero Boundary Conditions: The Challenge

Problem: $y''(x) = -1$ on $(0, 1)$ with $y(0) = 1$, $y(1) = 2$.

Analytical solution: $y(x) = -\frac{1}{2}x^2 + \frac{3}{2}x + 1$.

Soft BC enforcement:

$$\ell = \frac{1}{N} \sum_i r(x_i)^2 + \lambda_0 (\mathcal{N}_\theta(0) - 1)^2 + \lambda_1 (\mathcal{N}_\theta(1) - 2)^2$$

Issues with soft enforcement:

- ▶ BCs are only **approximately** satisfied
- ▶ Penalty weight λ requires **tuning**: too small $\Rightarrow$ BC violated; too large $\Rightarrow$ PDE ignored
- ▶ Boundary errors can **propagate** into the interior

$\Rightarrow$ **Motivates hard boundary enforcement** (Part III).

Part III

Soft vs. Hard Boundary Conditions

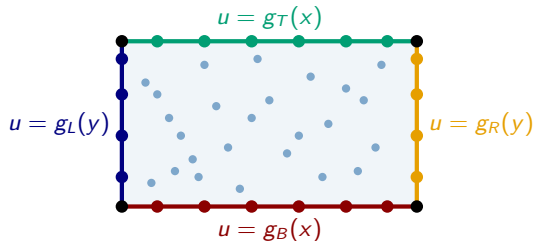
What Are Boundary Conditions?

1D: $y''(x) = -1$ on $(0, 1)$



Interior: enforce $y'' + 1 = 0$
Boundary: enforce $y = \text{prescribed value}$

2D: $\Delta u = f$ on $(0, 1)^2$

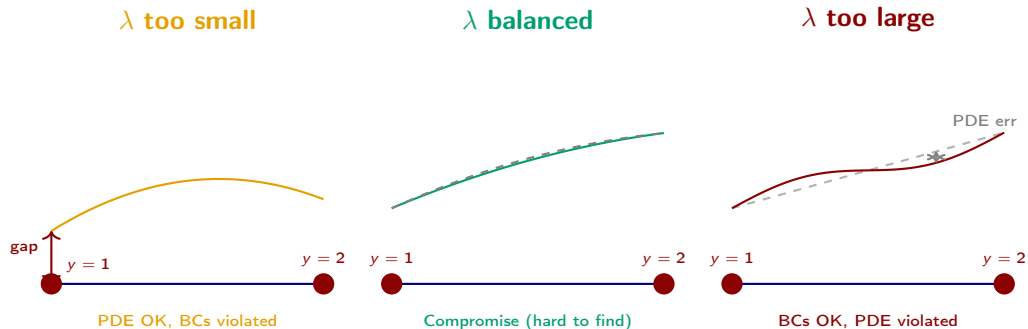


Two enforcement strategies

Soft: add BC violation as a penalty term in the loss.

Hard: build BCs into the network output by construction.

The Problem with Soft Boundary Conditions



Soft BCs require **careful balancing** of the penalty weight λ .
Too small: BCs violated. Too large: PDE ignored. Just right: hard to find *a priori*.

Hard BC Enforcement: Trial Solutions

Idea: construct a **trial solution** that satisfies BCs *by construction*:

$$\hat{y}(x) = A(x) + B(x) \cdot \mathcal{N}_\theta(x)$$

where:

- ▶ $A(x)$: **anchor function** satisfying the BCs exactly, e.g. $A(x) = 1 + x$
- ▶ $B(x)$: **mask function** vanishing at boundaries, e.g. $B(x) = x(1 - x)$
- ▶ $\mathcal{N}_\theta(x)$: free neural network (no BC constraints)

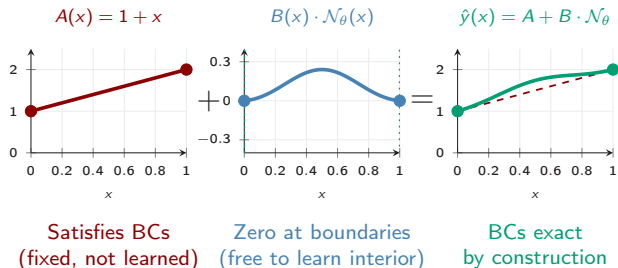
Verification:

$$\hat{y}(0) = A(0) + B(0) \cdot \mathcal{N}_\theta(0) = 1 + 0 = 1 \quad \checkmark$$

$$\hat{y}(1) = A(1) + B(1) \cdot \mathcal{N}_\theta(1) = 2 + 0 = 2 \quad \checkmark$$

$\Rightarrow$ BCs hold **identically**, only the PDE residual loss remains.

Trial Solution: Visual Decomposition



Hard BC: Implementation

```
def A(x): # Anchor: A(0)=1, A(1)=2
    return 1.0 + x
def B(x): # Mask: B(0)=B(1)=0
    return x * (1.0 - x)

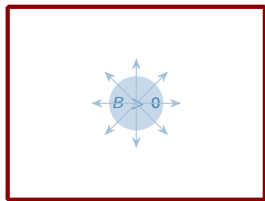
def y_trial(model, x): # Hard-BC trial solution
    return A(x) + B(x) * model(x)

def residual(model, x):
    x.requires_grad_(True)
    y = y_trial(model, x)
    dydx = torch.autograd.grad(y, x, ...)[0]
    d2ydx2 = torch.autograd.grad(dydx, x, ...)[0]
    return d2ydx2 + 1.0 # should be 0
```

Advantage: the loss has **only one term** (PDE residual), no λ to tune!

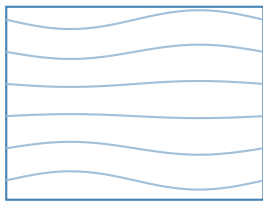
Hard BCs in 2D: Building the Trial Solution

Mask $B(x, y) = x(1-x)y(1-y)$



$B = 0$ on all edges

Network $\mathcal{N}_\theta(x, y)$



Free to take any value



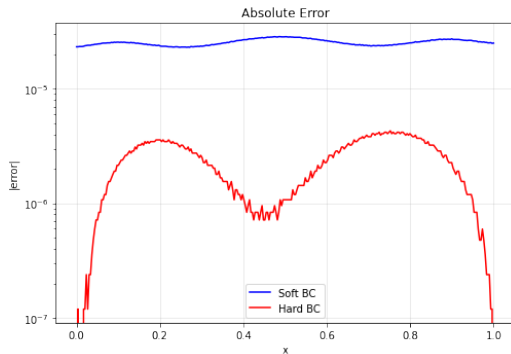
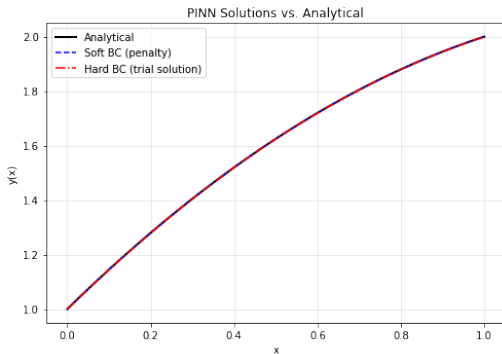
$B \cdot \mathcal{N}_\theta$: zero on $\partial\Omega$



Exactly zero on boundary

Then $\hat{u}(x, y) = \underbrace{A(x, y)}_{\text{matches BCs}} + \underbrace{B(x, y) \cdot \mathcal{N}_\theta(x, y)}_{\text{zero on } \partial\Omega}$ satisfies Dirichlet BCs **exactly**.

Soft vs. Hard BCs: Comparison



Both fit the solution (left). The difference is the error (right, log scale): the soft-BC run sits near 10^{-3} **including at the endpoints**, while the hard-BC run falls to 10^{-7} at the boundary, exact there by construction.

Notebook: 02_ODE_PINN_SoftVsHardBCs

Finger Exercise: Hard Boundary Conditions

Exercise (5 min)

Consider the same ODE $u''(x) = -1$ on $[0, 1]$ with $u(0) = u(1) = 0$.

1. Construct a **hard-BC trial solution** of the form $\hat{u}(x) = B(x) \cdot \mathcal{N}(x; \theta)$, where $B(x)$ enforces both BCs exactly. (Hint: B must vanish at $x = 0$ and $x = 1$.)
2. Verify that $\hat{u}(0) = \hat{u}(1) = 0$ for *any* network output $\mathcal{N}$.
3. What is the advantage of hard BCs over soft BCs?

Solution: Hard Boundary Conditions

Answer

1. $\hat{u}(x) = x(1 - x) \cdot \mathcal{N}(x; \theta)$. Since $B(x) = x(1 - x)$ vanishes at both endpoints, the BCs are satisfied exactly.
2. $\hat{u}(0) = 0 \cdot \mathcal{N}(0; \theta) = 0$. $\hat{u}(1) = 0 \cdot \mathcal{N}(1; \theta) = 0$. ✓
This holds regardless of what $\mathcal{N}$ outputs.
3. Hard BCs **eliminate the boundary loss term** entirely, the optimizer only needs to minimize the PDE residual. This reduces the number of hyperparameters (λ is gone) and often speeds up convergence.

Extension to 2D: Poisson Equation

Problem: $\Delta u = f(x, y)$ on $\Omega = (0, 1)^2$ with Dirichlet BCs.

Manufactured solution: $u^*(x, y) = x^2 + y + \sin(\pi x) \sin(\pi y)$.

Hard BCs via transfinite (Coons-patch) interpolation: the standard 4-side construction that matches each boundary function exactly along its edge:

$$\hat{u}(x, y) = A(x, y) + \underbrace{x(1-x)y(1-y)}_{B(x,y)} \cdot \mathcal{N}_\theta(x, y)$$

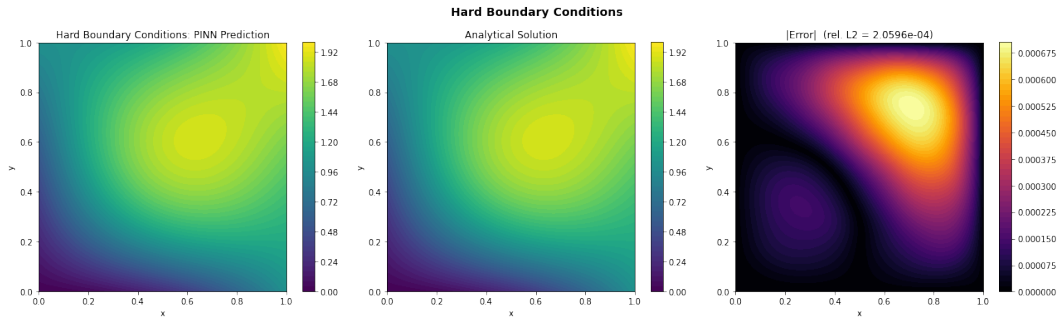
where $A(x, y)$ interpolates boundary data from all four sides:

$$A = (1-x) g_L(y) + x g_R(y) + (1-y) g_B(x) + y g_T(x) - \text{corners}$$

Laplacian via autograd:

$$\Delta \hat{u} = \frac{\partial^2 \hat{u}}{\partial x^2} + \frac{\partial^2 \hat{u}}{\partial y^2} \quad (\text{two chains of autograd.grad})$$

2D Poisson: Results



Hard-BC PINN, analytical solution, and absolute error. Relative L^2 error 2.06×10^{-4} against 1.36×10^{-3} for the tuned soft-penalty run: a **6.6×** improvement, at one fewer hyperparameter.

Notebook: 03_PDE_PINN_Poisson2D

Soft vs. Hard BCs: When to Use Which?

	Soft BCs	Hard BCs
Best for	Irregular domains, Robin or Neumann BCs	Simple domains, Dirichlet BCs, high accuracy
Pros	Flexible, easy to implement	Exact BCs, fewer hyperparams, better accuracy
Cons	λ tuning, approximate BCs	Needs analytical A , B ; harder for irregular domains

Practical recommendation

Start with **hard BCs** when the domain and boundary data allow it. Fall back to **soft BCs** for complex geometries or mixed boundary types.